PS: Smart Virtual Rehabilitation Companion

1. Context & Background

Modern fulfillment centers process thousands of orders daily. A major operational bottleneck in logistics is **suboptimal SKU placement (warehouse slotting)**. When high-demand items are located far from packing stations, or when heavy, bulky items are placed on upper shelf tiers without ergonomic consideration, order pickers spend a massive portion of their shift navigating inefficient travel routes.

Beyond travel inefficiencies, poor spatial layout causes underutilized warehouse volume, structural weight overload on rack beams, and elevated safety risks caused by storing incompatible product categories near one another.

Participants must build an end-to-end **Smart Warehouse Slotting & Optimization Engine** that dynamically maps SKUs to warehouse physical locations (bins, racks, shelf levels) based on historical demand patterns, 3D spatial dimensions, physical weight bounds, and operational handling constraints.

2. Core Problem & Functional Requirements

Your challenge is to develop an algorithmic system that ingests raw warehouse layout topographies, SKU physical metadata, and order transactional logs, processes them through optimization logic, and produces a fully feasible layout assignment alongside explicit quantitative proof of efficiency gains over an unoptimized baseline layout.

Key System Requirements:

1. **Rule-Based & Physical Constraint Enforcement:**

    - **Volumetric & Spatial Bounds:** Match SKU 3D dimensions (width, height, depth) and total volume against target bin dimensions and structural capacities without exceeding limits.
    - **Structural Load & Weight Rating:** Ensure total assigned weight per shelf level and rack frame strictly adheres to maximum safe load bearing limits.
    - **Category & Hazard Compatibility:** Enforce physical separation matrix rules (e.g., keeping hazardous materials isolated from standard inventory, preventing fragile items from being placed in high-risk zones, and honoring vertical shelf height limits for heavy SKUs).
2. **Demand-Aware & Frequency-Driven Optimization:**
    - **Velocity Profiling (ABC Analysis):** Process historical order streams to extract item demand frequency and velocity. Map high-velocity (Class A) SKUs to prime accessibility zones near Input/Output (I/O) dispatch areas.
    - **Order Affinity & Co-Occurrence:** Mine basket data for item co-occurrence (products frequently ordered together in single orders) and cluster them into adjacent pick-faces to minimize multi-item pick path traversal.
    - **Ergonomic & Replenishment Efficiency:** Consider golden-zone picking heights (chest-to-waist levels) for high-frequency picks to streamline worker handling.
3. **Comparative Evaluation & Impact Engine:**
    - Run comparative simulations between an initial/random baseline SKU distribution and your optimized layout.
    - Compute concrete metric improvements, focusing on total pick path travel distance reduction, overall volumetric cube utilization (fill rate %), pick route completion time, and zero-tolerance safety rule compliance.

3. Dataset Discovery, Acquisition & Exploratory Data Analysis (EDA) Mandate

> **Important:** No pre-packaged or ready-made warehouse dataset will be provided by the organizers. **Participants must search for, identify, acquire, synthesize, or curate their own dataset pipeline to solve and validate this problem statement.**

To build a fully functional prototype, your team is required to explore open-data repositories, public domain e-commerce datasets, or industrial benchmarks, and assemble a complete data pipeline consisting of three primary vectors:

Core Data Vectors to Sourcing & Synthesizing:

- **Order & Demand Data:** Sourced from real-world open transaction datasets (e.g., *Instacart Market Basket Analysis*, *Kaggle Online Retail II*, *Olist Brazilian E-Commerce*, or custom synthetic transaction generators). Must contain fields such as `Order_ID`, `SKU_ID`, `Quantity`, and `Timestamp`.
- **SKU Physical Master Data:** Physical dimensions and attributes for each item, including `SKU_ID`, `Width`, `Height`, `Depth`, `Weight`, `Category`, `Hazard_Class`, and `Handling_Type`.
- **Warehouse Topological Grid Data:** A structured representation of the physical warehouse environment, detailing `Location_ID`, `Zone`, `Aisle_Number`, `Rack_ID`, `Bay`, `Shelf_Level`, `X_Y_Z_Coordinates`, `Max_Weight_Capacity`, `Volumetric_Capacity`, and `Distance_to_Dispatch_Depot`.

Required Exploratory Data Analysis (EDA):

Before feeding data into your allocation engine, your submission must document a thorough exploratory data analysis phase covering:

- **Demand Velocity Profiling:** Perform Pareto distribution (80/20 rule) analysis to categorize items into Class A (fast-moving), Class B (medium-moving), and Class C (slow-moving) tiers.

- **Affinity & Co-Occurrence Matrix:** Mine transaction logs using market basket analysis (e.g., Apriori or association scoring) to identify product pairs and clusters frequently appearing in the same order batch.
- **Volumetric Density & Space Profiling:** Analyze product volume distributions against bin capacity classes to uncover spatial underutilization, volume mismatch bottlenecks, and density opportunities.

4. Expected Deliverables

1. **Working Allocation Prototype:** A functional software system/tool that accepts warehouse configurations and SKU demand datasets, runs allocation algorithms, and outputs feasible, constraint-compliant storage assignments.
2. **Interactive Optimization Dashboard:** A visualization interface showcasing layout visualizers (heatmaps/grid layouts of storage locations), pick-path travel simulations, and performance metrics (Baseline vs. Optimized).
3. **Technical & Algorithmic Documentation:** Clear documentation outlining system architecture, constraint handling logic, computational complexity, EDA insights, and trade-off analysis.